



Present Status Of Quantum Based Superfast Computer

This article is a review of the present status of quantum computers, which can solve certain problems much faster than a classical computer. Their future is now much more exciting and awaited



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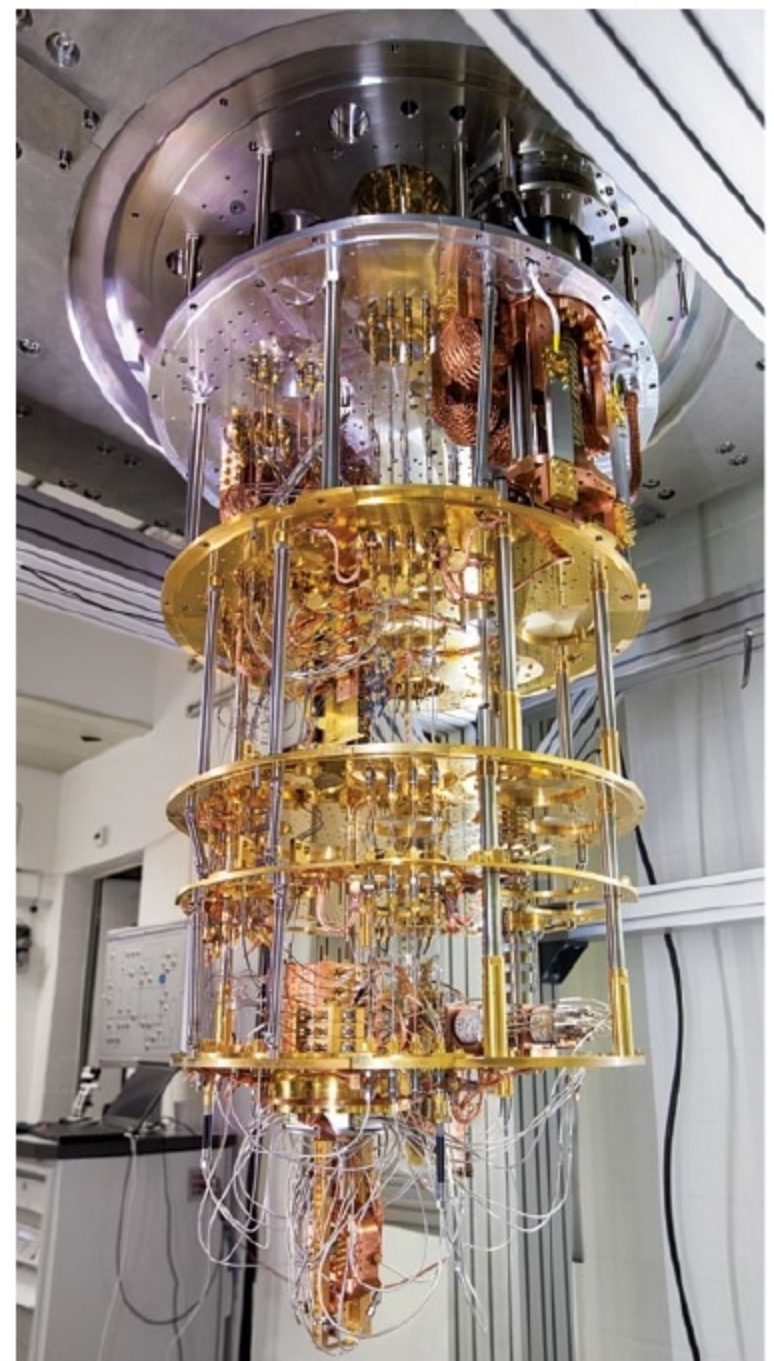
A computer is a device that takes an input, carries out a sequence of instructions, and produces an output. In a digital computer, these inputs, instructions and outputs are all sequences of 1s and 0s—individually called bits. A quantum computer is one that harnesses phenomena from quantum physics, the study of the behaviour of subatomic particles.

A quantum computer does the same thing as a classical computer, but it uses quantum bits or qubits. While a bit takes only one of the two values—1 or 0, a qubit uses the complex mathematics of quantum mechanics, providing a richer set of possibilities. As binary digits or bits are the basic units of information in classical computing, quantum bits or qubits are the basic unit of information in quantum computing.

Large-scale quantum computers would theoretically be able to solve certain problems much more quickly than any classical computer that uses even the best currently known algorithms, like integer factorisation. There exist quantum algorithms, such as Simon's algorithm, that run faster than any possible probabilistic classical algorithm. On the other hand, quantum computers may be able to efficiently solve problems which are not practically feasible on a classical computer.

The principle is that a particle can be in two different states at the same time—as with Schrodinger's imaginary cat, who was both alive and dead at the same time. This is known in the jargon as "superposition." Superposition is at the heart of quantum computing.

Ordinary computers work with bits that can be either on or off—coded as zero



Quantum computer based on superconducting qubits developed by IBM Research in Zürich, Switzerland. The qubits in the device shown here will be cooled to under 1 kelvin using a dilution refrigerator (Credit: en.wikipedia.org)

or one. But quantum computers work with qubits, which can have a value of 0, 1 or both. Thus, two qubits can represent four states simultaneously 00, 01, 10, 11. This means that a quantum computer would be much faster and efficient at some kind of computation than a classic computer,